

Data Flow Diagram

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Data Flow Diagram (DFD)

Introduction

A **Data Flow Diagram (DFD)** is a graphical representation of how data moves through a system. It illustrates the flow of information between users, system processes, machine learning models, and data storage components. The DFD helps visualize how user inputs are processed to generate a prediction while ensuring that data is managed efficiently throughout the application.

In the **Credit Card Approval Prediction using Machine Learning** project, the DFD demonstrates how applicant information entered through the web interface is processed by the Flask backend, passed to the trained Random Forest model for prediction, and finally displayed to the user as an approval or rejection result.

DFD Components :

External Entity

- Customer / Applicant

Processes

- User Registration / Input
- Data Validation
- Prediction Processing
- Result Generation

Data Stores

- Trained Machine Learning Model (credit_card_approval_model.pkl)

Output

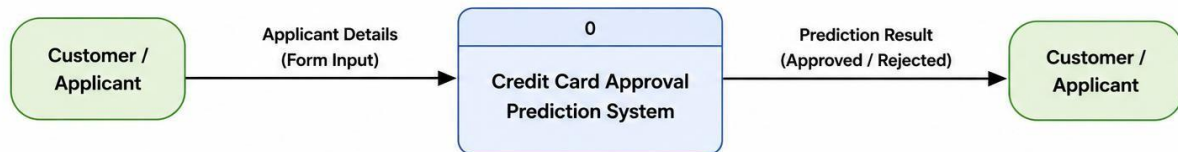
- Credit Card Approval / Rejection Result

Data Flow Diagram :

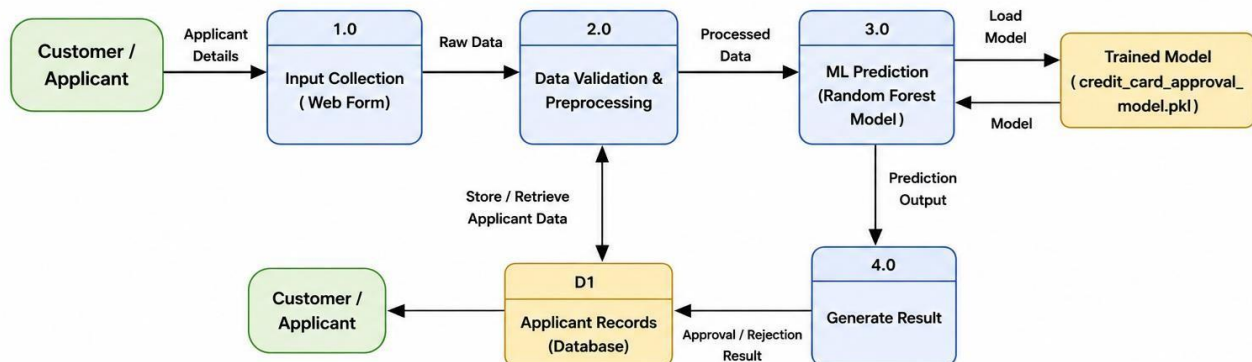
DATA FLOW DIAGRAM (DFD)

Project: Credit Card Approval Prediction using Machine Learning

LEVEL-0 DATA FLOW DIAGRAM



LEVEL-1 DATA FLOW DIAGRAM



Data Flow Explanation

The Data Flow Diagram (DFD) illustrates how applicant information moves through the Credit Card Approval Prediction system and how the machine learning model generates the final prediction.

Level-0 Data Flow Diagram

The Level-0 DFD provides a high-level overview of the system. The **Customer/Applicant** enters personal, demographic, and financial details through the web application. These details are sent to the **Credit Card Approval Prediction System**, which processes the input using the trained machine learning model. After processing, the system returns the prediction result to the customer, indicating whether the credit card application is **Approved** or **Rejected**.

Level-1 Data Flow Diagram

The Level-1 DFD explains the internal working of the prediction system in detail:

1. Input Collection (Web Form):

- The customer enters all the required personal, financial, and employment details through the web-based application form.
- The entered information is collected by the Flask backend.

2. Data Validation & Preprocessing:

- The submitted data is validated to ensure completeness and correctness.
- Categorical values are converted into numerical values using the same encoding technique used during model training.
- The processed data is prepared in the format required by the machine learning model.

3. Machine Learning Prediction (Random Forest Model):

- The preprocessed data is passed to the trained **Random Forest** model.
- The model is loaded from the stored file `credit_card_approval_model.pkl`.

- Based on the applicant's information, the model predicts whether the credit card application is likely to be approved or rejected.

4. Generate Result:

- The prediction generated by the machine learning model is processed by the Flask application.
- The final decision is displayed to the customer as either **Credit Card Approved** or **Credit Card Rejected**.

5. Applicant Records (Database):

- The system can optionally store applicant details and prediction results for future reference, reporting, or analysis.
- These records may also be retrieved whenever required.